

Project Development Phase

Smart Lender: AI-Powered Loan Approval Prediction System

Implementation Overview

The Smart Lender system was developed following the design created during the Project Design Phase. The implementation began with data preprocessing, including handling missing values, encoding categorical variables, balancing the dataset using SMOTE, and applying StandardScaler for feature scaling.

Multiple Machine Learning models were trained and evaluated using the processed dataset. The Gradient Boosting Classifier achieved the highest testing accuracy and was selected as the final prediction model. The trained model and scaler were saved using the Pickle library and integrated into a Flask web application. Users can enter applicant details through the web interface and receive an instant loan approval prediction.

Key Modules

- notebook/
 - Smart_Lender.ipynb (Data preprocessing, model training and evaluation)
 - model/
 - loan_model.pkl
 - scaler.pkl
 - templates/
 - home.html
 - input.html
 - output.html
 - static/
 - style.css
 - app.py
 - Flask application
 - Input validation
 - Loan predictio
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Model Results

The following Machine Learning algorithms were trained and evaluated:

- Decision Tree Classifier
- Random Forest Classifier
- K-Nearest Neighbors (KNN)
- Gradient Boosting Classifier

Among these models, the Gradient Boosting Classifier produced the highest testing accuracy and was selected as the final model for deployment.

Model Accuracy

Machine Learning Model	Training Accuracy	Testing Accuracy
Decision Tree	1.0000	0.7670
Random Forest	1.0000	0.7849
K-Nearest Neighbors (KNN)	0.8425	0.7061
Gradient Boosting Classifier	0.9204	0.7921